

Technical Document #683: Cybersecurity - Comprehensive Overview

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Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Encryption converts plaintext into ciphertext to protect data confidentiality.
- Zero trust architecture assumes no implicit trust and verifies every request.
- Incident response plans define how organizations respond to security incidents.